

Shaarav S

Backend Developer · CS Undergraduate · India
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ABOUT

CS undergrad focused on backend development. I build systems around authentication, caching, job queues, and REST API design — going beyond the API surface to understand the trade-offs behind each decision. I think in terms of data flow, bottlenecks, and failure modes under load. I also maintain a fully customized Arch-based Linux environment, tuning schedulers and system-level settings for low-latency workloads — the same systems thinking applied to infrastructure. Quick to pick up new tools, self-sufficient when researching, and looking for backend roles where I can take real ownership.

TECHNICAL SKILLS

Languages	JavaScript (Node.js), Python, SQL, Bash / Fish
Backend	Express.js, REST API Design, JWT Auth + Refresh Tokens, bcrypt, BullMQ, Rate Limiting
Databases	PostgreSQL, Redis (caching, queues, pub/sub)
Frontend	React, Vite, Tailwind CSS
Tools	Git, GitHub, Docker, Postman, npm, SSH
Environment	Linux (CachyOS / Arch), Fish shell, CLI-native workflow, system configuration
Interests	Network Security, AppSec, DevSecOps, System Design, OS Internals

PROJECTS

Secure REST API with JWT & Redis — Express.js · PostgreSQL · Redis · JWT · Railway

- Full auth system with JWT access tokens (15-min expiry) and rotating refresh tokens (7-day) — stateless but revocable.
- Redis-backed token blacklist and refresh token store; rotation invalidates old tokens on reuse.
- Redis caching layer on protected routes with per-key TTLs; cache invalidation on mutation.
- Rate limiting middleware (express-rate-limit + Redis store) to prevent brute-force and abuse.
- Deployed on Railway with environment-segregated secrets; full test coverage via Postman collection.

github.com/saltyip/jwt-redis-auth-api

Email Job Queue System — BullMQ · Redis · PostgreSQL · Node.js · Gmail SMTP · Railway

- Email job queue with BullMQ — producers enqueue jobs, workers process and dispatch via Gmail SMTP.
- PostgreSQL job logging for a persistent audit trail: job status, timestamps, and failure reasons survive restarts.
- Exponential backoff retry logic for transient SMTP failures and dead-letter handling.
- Real-time status endpoint to query job state; deployed and running on Railway.

github.com/saltyip/emailqueue

URL Shortener — Express.js · PostgreSQL · Redis · JWT · bcrypt

- Complete URL shortening service with user authentication (JWT + bcrypt), per-user link management, and redirect resolution.
- Base62 encoding implemented from scratch — understood the math and wrote it without a library.
- Redis caching layer for redirect lookups, reducing repeated database reads on high-traffic links.
- JWT middleware for protected routes; cleanly handled token expiry, invalid states, and error responses.

github.com/saltyip/urlshortnercongential

Nudge — Node.js · systemd · Fish / Bash / Zsh · chrono-node

- CLI tool for time-aware reminders that hooks directly into the shell — supports triggers like "in 2 hours" or "tomorrow 9am" via NLP parsing with chrono-node.
- Shell hooks (preexec/precmd pattern) enable --after-cmd triggers that fire when a specific command exits, caught via /proc/<pid> polling for instant detection.
- systemd user daemon checks every 5 seconds; reminders persist in ~/.local/share/nudge/nudges.json across reboots and sessions.
- Shell-agnostic — hook scripts ship for Fish, Bash, and Zsh with no changes to core logic.

github.com/saltyip/nudge

SIDE PROJECTS

Conversation Simulator — Node.js · Ollama

- Self-hosted AI conversation simulation with an imposter-style minigame — one model running multiple personality prompts locally via Ollama, hidden among human-style responses.

EDUCATION

B.Tech — Computer Science2024 – 2028

2nd Year · India

Coursework: Design & Analysis of Algorithms, Probability & Statistics, Operating Systems, Computer Networks

ADDITIONAL

- Pursuing cybersecurity through CTF competitions (PicoCTF) and platforms including TryHackMe and HackTheBox.
- Linux system optimization: CPU scheduler tuning (scx_lavd), boot time reduction, and low-latency configuration on Arch.
- Personal portfolio at shaarav.dev — interactive OS-style interface built from scratch with a pixel art cinematic intro.